

# Routing Scheme Evaluation Results

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# 1 Evaluation Setup

| Parameter          | Value                                                                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Seeds per combo    | 30 (seeds 1–30); values are mean makespan                                                                                                        |
| Schedulers         | HEFT: calibrated to runtime routing model<br>HEFT-1: direct-link BW; 0.001 MB/s for non-adjacent<br>HEFT-2: widest-path BW for all pairs (fixed) |
| Routing schemes    | W, S, SH, GS, GC, GB, GO, GSD, GSD-D                                                                                                             |
| Interference model | <code>csma_bianchi</code> (802.11ax, 5 GHz, 20 MHz)                                                                                              |
| RF parameters      | $P_{tx} = 20$ dBm, path-loss exponent $n = 3.0$                                                                                                  |
| Grid spacing       | 40 m; adjacent-link PHY rate $\approx 8.6$ MB/s                                                                                                  |
| Compute costs      | 150–1000 cu (heterogeneous, $\approx 6.7\times$ range)                                                                                           |
| Data sizes         | 2–30 MB (heterogeneous, $15\times$ range)                                                                                                        |
| Node capacities    | 80–300 cu/s (heterogeneous)                                                                                                                      |
| Total runs         | 3 schedulers $\times$ 9 routing $\times$ 4 experiments $\times$ 30 seeds = 3240                                                                  |

Table 1: Evaluation parameters.

**HEFT** (calibrated) passes the runtime routing model to HEFT’s pairwise rate matrix, so placement decisions are calibrated to whichever routing scheme runs at simulation time. Equals HEFT-2 when paired with `widest_path`.

**HEFT-1** pins 0.001 MB/s on all non-adjacent pairs regardless of runtime routing. Strongly biases placement toward same-node or direct-neighbour assignments.

**HEFT-2** always uses `WidestPathRouting` for pairwise BW estimates, regardless of the runtime routing scheme.

**SH** (Shortest Hop) minimises hop count, breaking ties by  $\sum 1/b_\ell$ . Added alongside S to test whether reducing relay count reduces `csma_bianchi` interference enough to offset lower bottleneck bandwidth.

## 2 Full Results Tables

Mean makespan in seconds, averaged over 30 seeds. **Bold green**: overall experiment winner. **Red**: worst entry in that column. †: best within that scheduler column.

## 2.1 $4 \times 4$ Grid, Small DAG (8 tasks, fork-join, 12 edges)

| Routing     | HEFT (s)                  | HEFT-1 (s)                | HEFT-2 (s)                |
|-------------|---------------------------|---------------------------|---------------------------|
| W           | <b>1814.518</b>           | 18.336                    | <b>1814.518</b>           |
| S           | 93.856                    | 18.336                    | 93.856                    |
| SH          | 93.856                    | 18.336                    | 93.856                    |
| GS          | 93.319                    | 18.336                    | 93.319                    |
| GC          | 93.319                    | 18.336                    | 93.319                    |
| GB          | 93.319                    | 18.336                    | 93.319                    |
| GO          | 93.319                    | 18.336                    | 93.319                    |
| GSD         | 140.483                   | 18.336                    | 140.483                   |
| GSD-D       | <b>91.134<sup>†</sup></b> | <b>18.035<sup>†</sup></b> | <b>91.134<sup>†</sup></b> |
| <i>Best</i> | <i>GSD-D 91.134</i>       | <i>GSD-D 18.035</i>       | <i>GSD-D 91.134</i>       |

Table 2:  $4 \times 4$  small. HEFT-1 co-locates all 8 tasks on one node; no transfers occur, so routing is irrelevant (all schemes tie at 18.336 s). GSD-D gains 0.3 s via deferral logic on same-node handoffs. HEFT and HEFT-2 are identical here (both use widest-path BW when routing=W; for other schemes the  $4 \times 4$  topology’s BW estimates converge). SH  $\equiv$  S throughout: grid paths that minimise  $\sum 1/b$  also minimise hops on this small topology.

## 2.2 $4 \times 4$ Grid, Large DAG (30 tasks, 5-stage pipeline, 48 edges)

| Routing     | HEFT (s)                   | HEFT-1 (s)                 | HEFT-2 (s)                 |
|-------------|----------------------------|----------------------------|----------------------------|
| W           | <b>2383.795</b>            | 497.788                    | <b>2383.795</b>            |
| S           | 1071.723                   | 292.425                    | 1071.723                   |
| SH          | 1071.723                   | 292.425                    | 1071.723                   |
| GS          | 1026.909                   | <b>292.091<sup>†</sup></b> | 1026.909                   |
| GC          | 1066.630                   | <b>292.091<sup>†</sup></b> | 1066.630                   |
| GB          | 1156.735                   | <b>292.091<sup>†</sup></b> | 1156.735                   |
| GO          | 1022.295                   | <b>292.091<sup>†</sup></b> | 1022.295                   |
| GSD         | <b>855.880<sup>†</sup></b> | 671.422                    | <b>855.880<sup>†</sup></b> |
| GSD-D       | 1037.424                   | 322.536                    | 1037.424                   |
| <i>Best</i> | <i>GSD 855.880</i>         | <i>GS/GC/GB/GO 292.091</i> | <i>GSD 855.880</i>         |

Table 3:  $4 \times 4$  large. HEFT and HEFT-2 are identical: with 30 tasks all paths converge regardless of routing calibration. Under HEFT-1, static greedy schemes (GS/GC/GB/GO) all tie at 292.091 s—41% vs W—by routing direct-neighbour transfers efficiently. Within the original HEFT, GSD wins at 855.880 s: calibrated placement spreads tasks and GSD’s runtime congestion avoidance pays off. SH  $\equiv$  S again: min-hop and min-delay paths coincide on this compact grid.

### 2.3 $7 \times 7$ Grid, Small DAG (8 tasks, fork-join, 12 edges)

| Routing     | HEFT (s)                  | HEFT-1 (s)                | HEFT-2 (s)                |
|-------------|---------------------------|---------------------------|---------------------------|
| W           | 1926.814                  | 24.608                    | 1926.814                  |
| S           | 93.902                    | 20.262                    | <b>93.696<sup>†</sup></b> |
| SH          | <b>93.819<sup>†</sup></b> | 20.262                    | 94.067                    |
| GS          | 1358.794                  | 20.262                    | 1309.776                  |
| GC          | 1135.942                  | 20.262                    | 1139.435                  |
| GB          | 1597.079                  | 20.262                    | 1545.814                  |
| GO          | 1494.147                  | 20.262                    | 1542.840                  |
| GSD         | 1393.943                  | 20.262                    | 1459.882                  |
| GSD-D       | 1512.787                  | <b>19.779<sup>†</sup></b> | 1338.502                  |
| <i>Best</i> | <i>SH 93.819</i>          | <i>GSD-D 19.779</i>       | <i>S 93.696</i>           |

Table 4:  $7 \times 7$  small. The only experiment where SH and S diverge under HEFT: SH (93.819 s) edges out S (93.902 s) by 0.08 s—a 0.1% difference, within noise. Under HEFT-2, S (93.696 s) wins. Under HEFT-1 all tasks are co-located; GSD-D wins at 19.779 s via deferral. Greedy static schemes under HEFT/HEFT-2 collapse to 1100–1600 s because calibrated placement spreads tasks but greedy routing cannot handle the congestion.

### 2.4 $7 \times 7$ Grid, Large DAG (60 tasks, 6-stage pipeline, 102 edges)

| Routing     | HEFT (s)                    | HEFT-1 (s)                 | HEFT-2 (s)                  |
|-------------|-----------------------------|----------------------------|-----------------------------|
| W           | 5695.953                    | 1318.190                   | 5657.358                    |
| S           | <b>2487.685<sup>†</sup></b> | 618.267                    | <b>2497.366<sup>†</sup></b> |
| SH          | 2536.137                    | 656.054                    | 2530.784                    |
| GS          | 4295.565                    | 660.732                    | 4432.067                    |
| GC          | 4431.863                    | 615.080                    | 4468.524                    |
| GB          | 4353.115                    | 624.117                    | 4511.408                    |
| GO          | 4439.492                    | <b>542.189<sup>†</sup></b> | 4361.533                    |
| GSD         | 4192.354                    | 1278.159                   | 4459.398                    |
| GSD-D       | 4436.056                    | 1556.583                   | 4474.458                    |
| <i>Best</i> | <i>S 2487.685</i>           | <i>GO 542.189</i>          | <i>S 2497.366</i>           |

Table 5:  $7 \times 7$  large. **S beats SH** under both HEFT (−2%) and HEFT-1 (−6%): on this topology, S’s 2-hop high-BW paths outperform SH’s 1-hop diagonals because bottleneck bandwidth dominates over relay interference. Under HEFT-1, GO wins at 542.189 s: overlap-based ordering best handles the dense parallel transfers in the 60-task pipeline, routing the most contested flows first. GSD and GSD-D are worse than W under HEFT-1 because compact placement leaves no beneficial alternate paths. Original HEFT calibrated to S delivers 2487.685 s—5.4× slower than HEFT-1+GO.

### 3 Cross-Experiment Comparison

#### 3.1 Overall Winners per Experiment

| Experiment         | HEFT best | (s)      | HEFT-1 best | (s)     | HEFT-2 best | (s)      |
|--------------------|-----------|----------|-------------|---------|-------------|----------|
| $4 \times 4$ small | GSD-D     | 91.134   | GSD-D       | 18.035  | GSD-D       | 91.134   |
| $4 \times 4$ large | GSD       | 855.880  | GS-GO       | 292.091 | GSD         | 855.880  |
| $7 \times 7$ small | SH        | 93.819   | GSD-D       | 19.779  | S           | 93.696   |
| $7 \times 7$ large | S         | 2487.685 | GO          | 542.189 | S           | 2497.366 |

Table 6: Best routing scheme per scheduler. HEFT-1 wins every experiment outright. HEFT and HEFT-2 are identical on  $4 \times 4$  (BW estimates converge on the compact grid).

#### 3.2 HEFT-1: Routing Rankings Across Experiments

| Routing | $4 \times 4$ S (s) | $4 \times 4$ L (s) | $7 \times 7$ S (s) | $7 \times 7$ L (s) | Wins |
|---------|--------------------|--------------------|--------------------|--------------------|------|
| W       | 18.336             | 497.788            | 24.608             | 1318.190           | 0    |
| S       | 18.336             | 292.425            | 20.262             | 618.267            | 0    |
| SH      | 18.336             | 292.425            | 20.262             | 656.054            | 0    |
| GS      | 18.336             | <b>292.091</b>     | 20.262             | 660.732            | 1    |
| GC      | 18.336             | <b>292.091</b>     | 20.262             | 615.080            | 1    |
| GB      | 18.336             | <b>292.091</b>     | 20.262             | 624.117            | 1    |
| GO      | 18.336             | <b>292.091</b>     | 20.262             | <b>542.189</b>     | 2    |
| GSD     | 18.336             | 671.422            | 20.262             | 1278.159           | 0    |
| GSD-D   | <b>18.035</b>      | 322.536            | <b>19.779</b>      | 1556.583           | 2    |

Table 7: HEFT-1 column winners. GO leads with 2 wins (the highest-value experiments). GSD-D wins 2 small-DAG experiments. SH never beats S under HEFT-1; S is consistently equal or better. GSD/GSD-D are worst on large grids: compact placement leaves no beneficial alternate paths for dynamic routing.

#### 3.3 HEFT: Routing Rankings Across Experiments

| Routing | $4 \times 4$ S (s) | $4 \times 4$ L (s) | $7 \times 7$ S (s) | $7 \times 7$ L (s) | Wins |
|---------|--------------------|--------------------|--------------------|--------------------|------|
| W       | <b>1814.518</b>    | <b>2383.795</b>    | <b>1926.814</b>    | <b>5695.953</b>    | 0    |
| S       | 93.856             | 1071.723           | 93.902             | <b>2487.685</b>    | 1    |
| SH      | 93.856             | 1071.723           | <b>93.819</b>      | 2536.137           | 1    |
| GS      | 93.319             | 1026.909           | 1358.794           | 4295.565           | 0    |
| GC      | 93.319             | 1066.630           | 1135.942           | 4431.863           | 0    |
| GB      | 93.319             | 1156.735           | 1597.079           | 4353.115           | 0    |
| GO      | 93.319             | <b>1022.295</b>    | 1494.147           | 4439.492           | 1    |
| GSD     | <b>91.134</b>      | <b>855.880</b>     | 1393.943           | 4192.354           | 2    |
| GSD-D   | <b>91.134</b>      | 1037.424           | 1512.787           | 4436.056           | 1    |

Table 8: Original HEFT (calibrated) column winners. GSD leads (2 wins), benefiting from calibrated placement that spreads tasks across the grid and runtime congestion avoidance. SH edges S by 0.08 s on  $7 \times 7$  small—within noise. S wins the  $7 \times 7$  large experiment by a clear margin over SH ( $-2\%$ ). W is always worst.

### 3.4 SH vs S: Does Hop-Count Routing Help?

| Experiment                             | S (s)           | SH (s)        | SH vs S (%) |
|----------------------------------------|-----------------|---------------|-------------|
| <i>Original HEFT (calibrated)</i>      |                 |               |             |
| 4 × 4 small                            | 93.856          | 93.856        | 0.0         |
| 4 × 4 large                            | 1071.723        | 1071.723      | 0.0         |
| 7 × 7 small                            | 93.902          | <b>93.819</b> | −0.1        |
| 7 × 7 large                            | <b>2487.685</b> | 2536.137      | +2.0        |
| <i>HEFT-1 (penalty)</i>                |                 |               |             |
| 4 × 4 small                            | 18.336          | 18.336        | 0.0         |
| 4 × 4 large                            | 292.425         | 292.425       | 0.0         |
| 7 × 7 small                            | 20.262          | 20.262        | 0.0         |
| 7 × 7 large                            | <b>618.267</b>  | 656.054       | +6.1        |
| <i>HEFT-2 (widest-path calibrated)</i> |                 |               |             |
| 4 × 4 small                            | 93.856          | 93.856        | 0.0         |
| 4 × 4 large                            | 1071.723        | 1071.723      | 0.0         |
| 7 × 7 small                            | <b>93.696</b>   | 94.067        | +0.4        |
| 7 × 7 large                            | <b>2497.366</b> | 2530.784      | +1.3        |

Table 9: SH vs S makespan comparison. **Bold**: lower of the two. SH never wins by a meaningful margin: the sole SH lead (7×7 small, HEFT) is 0.1%, within 30-seed averaging noise. S outperforms SH by 6% on 7 × 7 large under HEFT-1. Conclusion: **use S, not SH**, on these 40 m grids.

## 4 Analysis

### 4.1 Why HEFT-1 Dominates on csma\_bianchi Grids

HEFT-1’s 0.001 MB/s penalty forces all communicating tasks onto the same node or direct neighbours. On a csma\_bianchi grid, PHY rates drop steeply with distance and relay hops accumulate interference; co-location avoids both. HEFT and HEFT-2 spread tasks across the topology, where runtime interference is typically 5–10× worse than HEFT’s estimates.

### 4.2 Why SH Does Not Beat S on These Grids

The hypothesis was: fewer relay hops  $\Rightarrow$  fewer interfering transmitters  $\Rightarrow$  lower total interference  $\Rightarrow$  lower makespan. The empirical result contradicts this on the 40 m grids. The reason: on a grid with 8.6 MB/s cardinal links and lower-BW diagonal links at 56.6 m, the path that minimises  $\sum 1/b$  (S) typically uses 2-hop cardinal routes rather than 1-hop diagonal routes. The 2-hop path has higher bottleneck bandwidth (8.6 vs lower diagonal BW), which more than compensates for the second active link. SH’s 1-hop preference selects the diagonal when S would use a 2-hop cardinal path—SH gets fewer hops but lower throughput.

**When SH might win:** topologies where all alternate paths have the same bandwidth (equal-BW grids) but different hop counts, making the relay interference cost the binding constraint. On the heterogeneous-BW grids tested here, this condition is not met.

### 4.3 Original HEFT vs HEFT-2: When Does Calibration Matter?

On the  $4 \times 4$  grid, original HEFT and HEFT-2 produce identical results for all routing schemes. The topology is compact enough that different routing models return similar path bandwidths, so the pairwise rate matrices converge.

On the  $7 \times 7$  grid with the large DAG, calibration starts to matter: HEFT (calibrated to GSD) gives 4192 s for GSD-routing, while HEFT-2 gives 4460 s. The calibrated scheduler places tasks at nodes where GSD-style congestion avoidance is effective; HEFT-2’s widest-path calibration places them differently, and GSD cannot recover.

For S-routing on  $7 \times 7$  large, both perform similarly (2488 vs 2497 s) because S’s path bandwidths are close to widest-path bandwidths for most pairs.

### 4.4 GO Wins on $7 \times 7$ Large (HEFT-1)

Under HEFT-1, tasks are placed on direct neighbours. The 60-task pipeline creates many simultaneous transfers at each stage transition. GO (overlap-based ordering) routes the most contested flows first—flows whose time windows overlap with the most other flows get first pick of low-interference paths. On the denser  $7 \times 7$  grid (240 links, more routing diversity), this ordering extracts more benefit than start-time ordering (GS), delivering 542 s vs 661 s (−18%).

### 4.5 Key Takeaways

1. **HEFT-1 dominates on csma\_bianchi grids.** Co-location is optimal; HEFT and HEFT-2 spread tasks into heavy interference.
2. **Use GO with HEFT-1 on large grids.** Overlap-based ordering outperforms start-time (GS) and byte-size (GB) by routing the most contested flows first. On small grids, GSD-D is slightly better.
3. **SH does not improve over S** on heterogeneous-BW grids. Min-delay (S) beats min-hop (SH) because 2-hop high-BW paths deliver higher bottleneck bandwidth than 1-hop low-BW diagonals.
4. **W is always the worst routing scheme.** Widest-path routes flows through the highest-BW but also most-congested paths, maximising interference under all three schedulers.
5. **Original HEFT calibration matters on large topologies.** On  $7 \times 7$ , calibrated HEFT outperforms HEFT-2 for dynamic routing schemes (GSD) by correctly anticipating per-scheme path bandwidths.

## 5 Reproducing These Results

```
cd ncsim/  
python run_routing_eval.py  
# Output JSON: /tmp/ncsim_full_eval/eval_results.json  
# Runtime: ~590 s (3240 runs, 8 parallel workers)
```